Lab: Systems Around Us

Learning Goal: Identify and analyze systems, boundaries, and Markov Blankets in real-world examples.

Part 1: Identifying Systems

Instructions: For each of the following, identify: (a) the boundary, (b) internal components, (c) sensory inputs, (d) active outputs, (e) what the system "tries" to maintain, and (f) what happens when the boundary breaks down.

1. A goldfish in a bowl
2. A university
3. Your smartphone
4. A forest ecosystem
5. A social media platform

System	Boundary	Internal Components	Sensory Inputs	Active Outputs	Maintains	Breakdown Effect
Goldfish	Skin, scales	Organs, blood, brain	Eyes, lateral line, chemoreceptors	Fin movements, gill pumping	Body temperature, O_2 levels	Death
University	?	?	?	?	?	?
Smartphone	?	?	?	?	?	?
Forest	?	?	?	?	?	?
Social media	?	?	?	?	?	?

Complete the table for all five systems.

Write your response here

Part 2: Drawing Markov Blankets

Instructions: Draw a Markov Blanket diagram for each system below. Label: internal states (μ), external states (η), sensory states (s), and active states (a).

1. **A human cell**: Nucleus/organelles inside; extracellular environment outside; membrane receptors sense; membrane pumps and secretions act
2. **A country**: Government, citizens, institutions inside; other countries outside; diplomats, intelligence sense; military, trade, communication act
3. **Your brain**: Neural activity inside; body and external world outside; sensory nerves sense; motor nerves act

For each diagram, answer: Where is the boundary? What is conditionally independent of what?

Write your response here

Part 3: Homeostasis vs. Allostasis

Exercise: Classify each behavior as homeostatic (reactive) or allostatic (predictive), then explain your reasoning:

Behavior	Type	Reasoning
Sweating when you overheat	Homeostatic	Detects current deviation and corrects it
Eating breakfast before a long day	Allostatic	Anticipates future energy need
Putting on sunscreen before going outside	?	?
Pulling your hand from a hot stove	?	?
Studying for exams weeks in advance	?	?
Shivering when you're cold	?	?
A squirrel burying nuts in autumn	?	?
Your heart beating faster during exercise	?	?
A thermostat turning on the heat	?	?
Setting an alarm to wake up for class	?	?

Reflection question: Which type (homeostasis or allostasis) requires a more complex generative model? Why? How does this connect to the distinction between thermostats and brains?

Write your response here

Part 4: Systems at Multiple Scales

> **Learning Goal:** Recognize that Markov Blankets exist at multiple nested scales.

Exercise: Consider these nested systems:

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Atom → Molecule → Organelle → Cell → Tissue → Organ → Organism → Family →
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1. Pick any three consecutive levels. Draw the Markov Blanket at each level.
2. How does the "internal" at one level become the "external" at a lower level?
3. At which level(s) does something you'd recognize as "agency" emerge? Why?
4. Is the biosphere a system with a Markov Blanket? If so, what are its sensory and active states?

Write your response here

Part 5: The Zombie Thought Experiment

> **Learning Goal:** Think critically about the relationship between system maintenance and agency.

Exercise: Consider a philosophical zombie — a being physically identical to a human but with no subjective experience (no "inner life"). According to Active Inference, does a philosophical zombie:

1. Have a Markov Blanket? (Yes/No and why)
2. Minimize free energy? (Yes/No and why)
3. Have a generative model? (Yes/No and why)
4. Qualify as an agent? (Yes/No and why)
5. What does your answer to these questions reveal about the relationship between Active Inference and consciousness?

Write a 200-word reflection on what this thought experiment reveals about the limits and strengths of the Active Inference framework.

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	System identification	Boundaries, inputs, outputs
2	Formal modeling	Markov Blanket notation
3	Classification	Homeostasis vs. allostasis
4	Scale thinking	Nested Markov Blankets
5	Philosophical reasoning	Systems, agency, and consciousness